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**SMART WATER DISTRIBUTION AND LEAK DETECTION
MANAGEMENT SYSTEM USING MULTI AGENT
REINFORCEMENT LEARNING
A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

MLA0301-REINFORCEMENT LEARNING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE)**

Submitted by

ALLAPAREDDY JAHNAVI (192472214)

Under the Supervision of

Dr. G.A. SENTHIL

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “SMART WATER DISTRIBUTION AND LEAK DETECTION MANAGEMENT SYSTEM USING MULTI AGENT REINFORCEMENT LEARNING” has been carried out by ALLAPAREDDY JAHNAVI (192472214) under the supervision of Dr. G.A.SENTHIL and is submitted in partial fulfilment of the requirements for the current semester of the B.E COMPUTER SCIENCE AND ENGINEERING program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

G.A. SENTHIL

PROFESSOR

Department of CLOUD COMPUTING
**SIMATS Engineering,
SIMATS, Chennai – 602105.**

COURSE FACULTY

G.A. SENTHIL

Professor

Department of cloud computing
**SIMATS Engineering,
SIMATS, Chennai – 602105**

DECLARATION

I, ALLAPAREDDY JAHNAVI (192472214) of the COMPUTER SCIENCE AND ENGINEERING (AI), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “SMART WATER DISTRIBUTION AND LEAK DETECTION MANAGEMENT SYSTEM USING MULTI AGENT REINFORCEMENT LEARNING” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

Individual Contribution Statement

Student	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution (%)
ALLAPAREDDY JAHNAVI	Module 1: Water Distribution Monitoring and Data Management	Data collection and monitoring system	Data validation and monitoring analysis	Module 1 documentation	34%
K. NEEHA	Module 2: Intelligent Leak Detection and Fault Analysis	Leak detection and fault analysis model	Leak testing and performance evaluation	Module 2 documentation	33%
K. KAVYA	Module 3: Multi-Agent Reinforcement Learning-Based Water Distribution	RL agent development and optimization	Agent testing and result analysis	Module 3 documentation	33%

ABSTRACT

Water distribution networks are essential for supplying water efficiently to residential, commercial, and industrial areas. However, conventional water distribution systems often face challenges such as uneven water supply, pressure variations, unexpected demand, and water losses caused by leakage and faults. Continuous monitoring of network conditions is therefore necessary to improve operational efficiency, reduce water wastage, and support sustainable water management. The proposed project, “Sustainable Multi-Agent Reinforcement Learning Framework for Smart Water Distribution and Real-Time Leak Detection,” addresses these challenges through an intelligent, data-driven approach.

The framework consists of three major modules: Water Distribution Monitoring and Data Management, Intelligent Leak Detection and Fault Analysis, and Multi-Agent Reinforcement Learning-Based Water Distribution. The first module focuses on collecting, organizing, preprocessing, and monitoring important water-network parameters such as flow rate, pressure, tank level, water demand, consumption, and water loss. The collected data is processed to identify changes in normal operating conditions and is presented through a monitoring interface for real-time observation. The second module uses the monitored parameters to identify abnormal conditions, leaks, and faults. The third module applies Multi-Agent Reinforcement Learning to optimize water distribution decisions by considering changing demand, pressure, and network conditions.

The implementation demonstrates that continuous monitoring and structured water-network data management can provide reliable inputs for leak detection and intelligent distribution optimization. The monitoring module enables visualization of network conditions and supports the identification of abnormal flow and pressure patterns. In particular, Module 1: Water Distribution Monitoring and Data Management forms the foundation of the proposed system by providing organized and continuously updated network data to the subsequent modules. The project concludes that integrating real-time monitoring, intelligent fault analysis, and Multi-Agent Reinforcement Learning can improve the efficiency, reliability, and sustainability of smart water distribution systems.

Keywords

Smart Water Distribution, Multi-Agent Reinforcement Learning, Real-Time Monitoring, Leak Detection, Water Network Data Management, Sustainable Water Management.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
Q	Flow Rate	L/min
P	Pressure	bar
D	Water Demand	L/min
C	Water Consumption	L
WS	Water Supplied	L
WL	Water Loss	L
WL%	Water-Loss Percentage	%
TL	Tank Level	%
t	Monitoring Time Step	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Water distribution systems are critical infrastructure used to supply water to residential, commercial, agricultural, and industrial users. A typical distribution network consists of reservoirs, storage tanks, pumps, pipelines, valves, and distribution zones. The efficient operation of such a network depends on maintaining suitable pressure, ensuring adequate flow, satisfying changing water demand, and minimizing water losses.

Traditional water distribution systems generally depend on periodic inspection, manually collected measurements, and fixed operational strategies. These approaches may not provide sufficient information about rapidly changing network conditions. Variations in water demand, pressure fluctuations, pipe failures, and leakage can result in inefficient distribution and significant water loss. Therefore, continuous monitoring and intelligent decision-making are important for developing sustainable and reliable water distribution systems.

The proposed project, “Sustainable Multi-Agent Reinforcement Learning Framework for Smart Water Distribution and Real-Time Leak Detection,” integrates real-time monitoring, data management, leak and fault analysis, and Multi-Agent Reinforcement Learning (MARL). The framework is divided into three modules: Module 1 – Water Distribution Monitoring and Data Management, Module 2 – Intelligent Leak Detection and Fault Analysis, and Module 3 – Multi-Agent Reinforcement Learning-Based Water Distribution.

Module 1 forms the foundation of the complete framework. It focuses on collecting and managing important water-network parameters such as flow rate, pressure, tank level, water demand, water consumption, and water loss. The collected data is processed, organized, and monitored to understand the current operating condition of the network. The monitoring information is then made available to the leak detection and optimization modules.

Thus, the project combines continuous network monitoring with intelligent analysis and learning-based control to improve water distribution efficiency, reduce losses, and support sustainable water resource management.

1.2 Need for the Project

Water loss and inefficient distribution are major engineering challenges in modern water networks. Leakage may occur because of damaged pipelines, faulty valves, excessive pressure,

aging infrastructure, or unexpected changes in network conditions. If these conditions are not identified early, they can increase water wastage, operational costs, and service disruptions.

The problem needs to be addressed because:

- Water is a limited and valuable natural resource, making efficient distribution essential.
- Leakage can cause significant loss of treated water before it reaches consumers.
- Pressure variations can affect the reliability and quality of water supply.
- Changing consumer demand makes fixed distribution strategies less effective.
- Manual monitoring may not provide continuous information about network conditions.
- Water utilities require timely information to identify abnormal network behavior and take corrective action.

The problem affects water utility operators, maintenance teams, local authorities, industries, residential consumers, and the environment. Operators require accurate network information for decision-making, while consumers depend on reliable water supply with adequate pressure. Existing systems may monitor individual parameters but often lack an integrated framework that combines real-time data monitoring, data management, leak detection, and intelligent distribution optimization. Fixed-rule approaches also have limited adaptability when network conditions change.

From an engineering perspective, continuous monitoring provides the necessary operational data for detecting abnormal conditions and making informed control decisions. Therefore, Module 1 is particularly significant because the quality, organization, and availability of monitoring data directly influence the effectiveness of the subsequent leak detection and reinforcement learning modules.

1.3 Problem Statement

Water distribution networks experience continuously changing conditions due to variations in water demand, flow rate, pressure, storage level, consumption, leakage, and network faults. Conventional monitoring and fixed operational approaches may not respond effectively to these interacting factors, resulting in water losses, pressure imbalance, inefficient resource utilization, and unreliable distribution.

The engineering problem is to develop an integrated smart water distribution framework capable of continuously collecting, managing, and monitoring water-network parameters, identifying abnormal operating conditions and leakage, and supporting optimized water distribution through Multi-Agent Reinforcement Learning while satisfying practical constraints

such as water availability, pressure requirements, changing demand, network capacity, and operational safety.

Within this overall problem, the individual contribution focuses primarily on Module 1: Water Distribution Monitoring and Data Management, which must reliably collect, preprocess, organize, and visualize network data to provide accurate inputs for leak detection and intelligent distribution optimization.

1.4 Project Aim

The aim of this project is to develop a sustainable smart water distribution framework that combines real-time water-network monitoring, intelligent leak and fault detection, and Multi-Agent Reinforcement Learning to improve distribution efficiency, reduce water loss, and support reliable water resource management.

The individual module aims to develop a reliable water distribution monitoring and data management system that continuously tracks important network parameters and provides organized data for subsequent analysis and optimization.

1.5 Project Objectives

The objectives of the project are:

1. To design a smart water distribution framework integrating monitoring, leak detection, and intelligent distribution optimization.
2. To develop a water-network data management system for collecting and organizing flow, pressure, tank level, demand, consumption, and water-loss data.
3. To implement a real-time monitoring module for observing changing water-network conditions through numerical indicators and graphical visualization.
4. To preprocess and analyze collected network data to identify normal and abnormal operating patterns.
5. To develop an intelligent leak and fault analysis module using monitored flow and pressure variations.
6. To model the water distribution problem as a multi-agent decision-making environment considering changing demand and network conditions.
7. To implement Multi-Agent Reinforcement Learning for improving water distribution decisions while maintaining required operational conditions.
8. To evaluate the framework using parameters such as water loss, flow consistency, pressure conditions, demand satisfaction, and distribution efficiency.

Individual Objectives – Module 1

Since Module 1 is the individual student contribution, the specific objectives are:

1. To collect and organize water distribution network parameters such as flow rate, pressure, tank level, demand, consumption, and water loss.
2. To preprocess and manage the collected data for reliable downstream analysis.
3. To develop a real-time monitoring interface for displaying the current condition of the water network.
4. To visualize parameter variations using suitable metrics, tables, and graphs.
5. To identify abnormal changes in monitored parameters that may indicate potential network problems.
6. To provide structured monitoring data as input to the leak detection and reinforcement learning modules.

1.6 Scope

Included in the Project

The project includes:

- Monitoring of water distribution network parameters.
- Collection and management of simulated or available water-network data.
- Monitoring of flow rate, pressure, tank level, water demand, consumption, and water loss.
- Data preprocessing and organization.
- Real-time or step-wise visualization of network conditions.
- Identification of abnormal flow and pressure behavior.
- Intelligent leak and fault analysis.
- Modeling of water distribution as a multi-agent environment.
- Application of Multi-Agent Reinforcement Learning for distribution optimization.
- Evaluation of water distribution efficiency and water-loss reduction.
- Development of a prototype monitoring and decision-support framework.

Individual Scope – Module 1

The individual work is primarily limited to:

- Water network data collection.
- Data preprocessing and management.
- Monitoring of network parameters.
- Real-time dashboard-based visualization.
- Analysis of flow, pressure, demand, tank level, consumption, and water-loss trends.

- Generation of structured data required by Modules 2 and 3.

Excluded from the Scope

The project does not include:

- Physical installation of water pipelines.
- Large-scale deployment of physical sensors.
- Replacement or repair of real-world pipelines.
- Direct control of municipal water infrastructure.
- Development of a complete commercial water utility management system.
- Guaranteeing performance under every possible real-world network configuration.

Assumptions

The following assumptions are made:

- Water-network measurements or simulated values are available at regular intervals.
- Flow and pressure measurements are sufficiently accurate for monitoring purposes.
- The network contains defined distribution zones and storage facilities.
- Demand values may vary with time.
- Network parameters can be represented digitally for analysis.
- The reinforcement learning environment receives reliable monitoring information.

Boundary Conditions

The framework operates within defined limits such as:

- Available water/storage capacity.
- Pipeline flow capacity.
- Acceptable pressure range.
- Distribution demand requirements.
- Pump/valve operating limits.
- Defined monitoring time intervals.
- Available network data and simulation conditions.

1.7 Expected Engineering Outcome

The expected outcome is a prototype intelligent water distribution monitoring and optimization framework capable of integrating water-network monitoring, leak detection, and reinforcement-learning-based distribution.

The major engineering outcomes include:

1. Smart Monitoring System

A monitoring interface that displays important network parameters such as:

- Flow rate
- Pressure
- Tank level
- Water demand
- Water consumption
- Water loss

2. Water Network Data Management Process

A structured process for collecting, preprocessing, storing, and presenting water-network data for further analysis.

3. Intelligent Leak Detection Support

The monitored data will provide the required flow and pressure information for identifying abnormal conditions and potential leakage.

4. Reinforcement Learning Environment

A computational environment representing the water distribution network, where multiple agents can make distribution-related decisions.

5. Multi-Agent Optimization Model

A MARL-based model capable of learning suitable distribution actions according to changing demand and network conditions.

6. Performance Evaluation

The framework will be evaluated using measures such as water loss, pressure stability, demand satisfaction, flow distribution, and overall distribution efficiency.

Individual Student Engineering Outcome – Module 1

The primary individual outcome is a functional Water Distribution Monitoring and Data Management module that acts as the data foundation of the complete system. It collects and processes network information, presents real-time operational conditions through a monitoring dashboard, tracks parameter changes, and supplies structured information to Module 2 (Leak Detection and Fault Analysis) and Module 3 (MARL-Based Distribution Optimization).

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focused on smart water distribution, network monitoring, leak detection, data management, and Reinforcement Learning. Research papers, technical sources, and existing technologies were reviewed, with emphasis on IoT monitoring, hydraulic simulation, machine learning, and MARL. EPANET was also reviewed as a water-network simulation tool for analyzing flow, pressure, tanks, pumps, and valves.

2.2 Review of Existing Work

2.2.1 Conventional Water Distribution Monitoring

Traditional systems use flow meters, pressure sensors, and tank-level sensors. They provide useful network information but may depend on periodic monitoring and can lack centralized real-time visibility.

Relevance: Module 1 combines major network parameters into a centralized monitoring system.

2.2.2 EPANET-Based Hydraulic Modeling

EPANET models pipes, junctions, pumps, valves, tanks, and reservoirs and provides hydraulic results such as flow and pressure.

Limitation: It is mainly a simulation tool and does not itself provide complete real-time monitoring, leak detection, and MARL control.

Relevance: It provides a useful basis for network simulation.

2.2.3 IoT-Based Smart Water Monitoring

IoT systems use connected sensors for continuous water-network monitoring and remote data collection. They improve visibility but face challenges such as sensor cost, noise, communication problems, and data quality.

Relevance: Module 1 follows this monitoring concept using structured network data.

2.2.4 Machine Learning for Water Networks

Machine learning can analyze network data for demand prediction, infrastructure analysis, and abnormal-condition detection. However, performance depends on data quality and network conditions.

Relevance: Module 1 provides the structured data required for intelligent analysis.

2.2.5 Leak Detection

Leak detection methods analyze flow and pressure changes to identify abnormal conditions. Their accuracy can be affected by sensor placement, noise, demand variations, and leak location.

Relevance: Module 1 supplies flow and pressure data to Module 2.

2.2.6 Reinforcement Learning and MARL

Reinforcement Learning learns suitable actions through interaction with an environment. MARL extends this to multiple agents and is suitable for distributed water-network optimization.

Relevance: Module 3 uses MARL, while Module 1 provides the network-state information required by the agents.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Conventional monitoring	Simple, familiar	Limited real-time visibility	Module 1 provides centralized monitoring
IoT monitoring	Continuous data collection	Cost, noise, communication issues	Supports Module 1 monitoring concept
EPANET	Strong hydraulic simulation	Mainly simulation-based	Supports network modeling
Rule-based detection	Simple and interpretable	Fixed thresholds	Used as a basic comparison
ML leak detection	Detects complex patterns	Requires quality data	Module 1 provides data to Module 2
Single-agent RL	Adaptive decisions	Less suitable for distributed networks	Foundation for MARL
MARL	Multi-agent coordination	Higher complexity	Used in Module 3
Proposed framework	Integrates monitoring, detection and optimization	Requires reliable data and integration	Complete smart-water framework

2.4 Research / Engineering Gap

Existing approaches often address monitoring, leak detection, simulation, or optimization separately. There is a need for an integrated framework that connects network monitoring → data management → leak detection → intelligent optimization.

For this project, the major gap addressed by Module 1 is the lack of a centralized and structured monitoring layer that combines flow, pressure, tank level, demand, consumption, and water loss and provides reliable network-state information to the subsequent modules.

2.5 Proposed Contribution

The proposed system integrates:

Module 1: Monitoring & Data Management



Module 2: Leak & Fault Analysis



Module 3: MARL-Based Optimization

Individual Contribution – Module 1

The primary individual contribution is the development of the Water Distribution Monitoring and Data Management module, which:

- Collects and organizes network data.
- Monitors flow, pressure, tank level, and demand.
- Calculates water consumption and water loss.
- Visualizes network conditions through a dashboard.
- Identifies abnormal parameter changes.
- Provides structured network-state data to Modules 2 and 3.

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CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder	Requirement
Water utility operator	View the current condition of the water distribution network through a centralized monitoring interface.
Network maintenance team	Identify abnormal flow, pressure, tank-level, and water-loss conditions at an early stage.
System administrator	Maintain organized and reliable network data for monitoring and analysis.
Water management authority	Reduce unnecessary water loss and improve distribution efficiency.
End users / consumers	Receive a reliable water supply with suitable pressure and adequate availability.
Project development team	Obtain structured monitoring data that can be supplied to leak-detection and MARL modules.

3.1.2 Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Collect water-network parameters	Functional	Flow, pressure, tank level, demand, consumption and water-loss values shall be recorded.
Store and organize monitoring data	Functional	Data shall be maintained in a structured tabular/digital format.
Display real-time/step-wise network status	Functional	Dashboard shall update network parameters for each monitoring interval.
Visualize parameter variations	Functional	Provide numerical indicators and time-series charts.
Identify abnormal parameter changes	Functional	Flag values exceeding predefined operating limits.

Provide data to subsequent modules	Functional	Structured output shall be available for leak analysis and MARL.
Monitoring accuracy	Non-Functional	System shall use validated input values and identify missing/invalid data.
Response time	Non-Functional	Dashboard updates should occur within a practical monitoring interval.
Reliability	Non-Functional	System should continue monitoring when individual non-critical data points are unavailable.
Usability	Non-Functional	Interface shall present important parameters clearly without requiring specialist software knowledge.
Security	Non-Functional	Access to stored/monitored data shall be controlled and unauthorized modification prevented.
Sustainability	Non-Functional	The system shall support reduction of water loss and unnecessary resource consumption.

3.2 Constraints & Applicable Standards

Only standards directly relevant to the proposed smart-water monitoring and control framework are considered.

Standard / Code	Requirement	How Applied in This Project
ISO 24516-1:2016 + Amendment 1:2025	Provides guidance for management of drinking-water distribution-network assets, including monitoring and performance review.	Used as a reference for monitoring network condition, performance indicators, and long-term management considerations.
ISO 24516-2:2019	Provides guidance for management of waterworks assets including pumping stations, reservoirs, tanks and metering infrastructure.	Relevant to the representation and monitoring of tanks, reservoirs, pumping-related infrastructure and measurement data.
IEC 62443-2-1:2024	Defines security-program requirements for industrial	Used as a cybersecurity reference for protecting monitoring/control

	automation and control-system asset owners.	data and considering access management.
IEC 62443-4-2:2019	Defines technical security requirements for industrial automation and control-system components, including identification, integrity, confidentiality and availability.	Used as a reference for secure handling of the monitoring application and its data interfaces.

Engineering Constraints

The major project constraints are:

- **Data constraint:** Sensor or simulated data may contain missing, noisy, or abnormal measurements.
- **Network constraint:** Flow and pressure must remain within defined operating limits.
- **Storage constraint:** Tank/reservoir levels cannot exceed defined capacity limits.
- **Demand constraint:** The system should attempt to satisfy changing water demand.
- **Computational constraint:** MARL training and simulation can require significant computational resources.
- **Cost constraint:** A prototype should minimize dependence on expensive physical sensing infrastructure.
- **Safety constraint:** The prototype should not directly operate real municipal pumps or valves without appropriate engineering validation and protection.
- **Cybersecurity constraint:** Monitoring data and control interfaces must be protected against unauthorized access or modification.

EPANET may be used as a hydraulic simulation reference or supporting tool because it models water distribution systems containing pipes, pumps, valves, tanks and reservoirs and supports analysis of operational conditions.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
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Flow-rate monitoring	Continuous/step-wise monitoring	L/min	High
Pressure monitoring	Monitor network pressure at defined points	bar	High
Tank level	Monitor available storage	%	High
Water demand	Track changing demand	L/min	High
Water consumption	Calculate/record consumption	L/min or L	Medium
Water loss	Estimate difference between supplied and consumed water	L/min or %	High
Monitoring interval	Regular update of network status	Time step	High
Data storage	Structured monitoring records	Records	High
Dashboard update	Display latest network condition	Seconds/time step	High
Anomaly indication	Flag abnormal operating conditions	Status	High
Data availability	Provide processed data to Modules 2 and 3	Dataset	High
System usability	Clear and understandable dashboard	Qualitative	Medium
Data security	Controlled access and protected records	Qualitative	High
Water-loss objective	Support reduction in avoidable losses	%	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Conventional Rule-Based Monitoring

A rule-based monitoring system uses predefined threshold values for flow, pressure, tank level, and water loss. It is simple and computationally inexpensive, but its response is dependent on manually selected thresholds and may not adapt effectively to changing network conditions.

Alternative 2: Machine-Learning-Based Monitoring

A machine-learning approach learns patterns from historical water-network data and can identify abnormal operating behavior. It can provide better adaptability than fixed rules, but its performance depends on data quality, training data, and model selection.

Alternative 3: Integrated Monitoring + MARL Framework

This approach combines structured network monitoring, intelligent leak/fault analysis, and Multi-Agent Reinforcement Learning. It provides a complete path from network-state observation to intelligent decision-making but has greater implementation and computational complexity.

Evaluation Matrix

Scoring: 1 = Poor, 3 = Moderate, 5 = Excellent

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3	4	5
Cost	15%	5	3	3
Safety	20%	4	4	4
Sustainability	20%	3	4	5
Reliability	15%	3	4	5
Weighted Score	100%	3.60 / 5	3.90 / 5	4.55 / 5

Selection

Alternative 3 achieves the highest weighted score because it provides the strongest combination of performance, sustainability, and reliability while supporting the complete project objective.

For the **individual contribution**, **Module 1 is the most important foundation of Alternative 3**, because the MARL and leak-detection components depend on reliable monitoring data.

3.5 Selected Design & Detailed Design

Justification for Final Design Selection

The integrated monitoring and intelligent framework was selected because it provides a complete engineering workflow rather than addressing only one aspect of water management. The evaluation in Section 3.4 gives the integrated approach the highest weighted score of 4.55/5.

The selected architecture separates the system into three connected modules:

Water Network → Module 1: Monitoring & Data Management → Module 2: Leak/Fault Analysis → Module 3: MARL Optimization

Architecture / System Flow

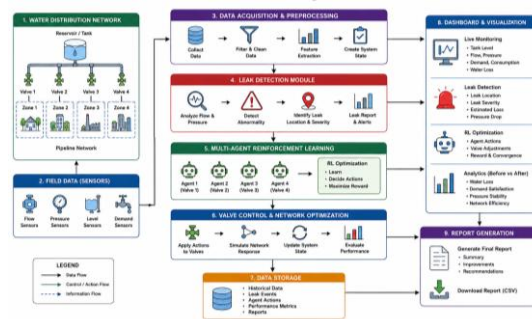


Figure 1: System Architecture

The system begins with water-network parameters such as flow, pressure, tank level, demand, consumption, and water loss. **Module 1 collects and manages these parameters and presents them through the monitoring interface.** The processed data is then supplied to Module 2 for abnormality and leak analysis and to Module 3 for intelligent distribution optimization.

Key Engineering Calculations

The following basic relationships are used in Module 1:

Water Loss:

$$\text{Water Loss} = \text{Water Supplied} - \text{Water Consumed}$$

Water-Loss Percentage:

$$Water\ Loss(\%) = \frac{Water\ Supplied - Water\ Consumed}{Water\ Supplied} \times 100$$

Demand Satisfaction:

$$Demand\ Satisfaction(\%) = \frac{Water\ Delivered}{Water\ Demand} \times 100$$

Tank Level:

$$Tank\ Level(\%) = \frac{Current\ Volume}{Maximum\ Tank\ Capacity} \times 100$$

These calculations allow Module 1 to convert raw network measurements into meaningful monitoring indicators.

Systems Approach

Module 1 receives network measurements and converts them into structured monitoring information. The processed network state is passed to Module 2, where abnormal flow and pressure conditions can be analyzed for possible leakage or faults. Module 3 uses the network state and analysis information to support intelligent distribution decisions through MARL. Therefore, the modules are interdependent, with **Module 1 acting as the primary data-input layer for the complete framework.**

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Monitoring method	Manual/periodic monitoring	Low complexity and cost	Limited real-time visibility	Automated/step-wise monitoring selected because continuous network-

				state information is required.
Data representation	Separate parameter records	Simple implementation	Difficult to analyze relationships	Integrated structured dataset selected to combine flow, pressure, demand, tank level and loss.
Fault identification	Fixed thresholds	Easy to understand	Less adaptable to changing conditions	Intelligent analysis selected because network conditions vary with demand and operating state.
Optimization	Fixed control rules	Simple and predictable	Limited adaptability	MARL selected because multiple interacting zones and changing conditions require adaptive decision-making.
Monitoring interface	Static tables	Easy to implement	Poor visualization of trends	Dashboard with metrics and charts selected for improved operational visibility.
Data source	Physical sensors only	Real-world measurements	Higher cost and deployment complexity	Simulation/available data for prototype selected , with physical sensors considered for future deployment.

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics, Health & Safety, and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary water loss and improve distribution efficiency.	Water supplied, consumed, lost, tank level and demand are monitored. ISO 24516-1 addresses management of drinking-water distribution assets and monitoring/performance review.	Monitoring water-loss indicators was made a key Module 1 requirement.
Sustainability	Avoid unnecessary pumping and distribution.	Flow, pressure, demand and storage conditions are tracked and supplied to the optimization module.	The system supports informed distribution decisions and provides data for MARL optimization.
Ethics	Ensure monitoring data is used responsibly.	Data is treated as operational system information and access is controlled.	Unauthorized modification or inappropriate use of network data is avoided.
Ethics	Avoid claiming prototype results as real-world infrastructure performance.	Results are obtained under defined simulation/prototype conditions.	Results are reported with their experimental assumptions and limitations.
Health & Safety	Maintain safe pressure and storage operating limits.	Pressure and tank-level values are continuously monitored.	Unsafe operating conditions can be flagged rather than blindly acted upon.
Health & Safety	Prevent unvalidated AI actions from directly	Prototype is evaluated computationally before real-world deployment.	MARL remains a decision-support/prototype component unless

	controlling infrastructure.		independently validated.
Security	Protect monitoring and control data.	IEC 62443 provides cybersecurity requirements for industrial automation and control systems.	Access control, data integrity, secure interfaces and controlled operation are considered.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect sensor/simulation data	Medium	High	High	Validate inputs and apply data-quality checks.	Medium
Missing network data	Medium	Medium	Medium	Handle missing values and retain previous valid states where appropriate.	Low
Abnormal pressure condition	Medium	High	High	Define acceptable pressure limits and generate warnings.	Medium
Excessive water loss	Medium	High	High	Monitor supplied-versus-consumed water and flag increasing losses.	Medium
Tank overflow/depletion	Low– Medium	High	Medium	Monitor tank level and enforce operating boundaries.	Low
Incorrect leak classification	Medium	High	High	Combine multiple parameters and	Medium

				validate against known scenarios.	
MARL produces unsuitable action	Medium	High	High	Apply action constraints and evaluate in simulation before deployment.	Medium
Communication/data-transfer failure	Medium	Medium	Medium	Use structured data handling and fault-tolerant processing.	Low
Unauthorized access	Low–Medium	High	Medium	Apply authentication, access control and secure data handling.	Low
Data manipulation	Low	High	Medium	Protect stored data and validate incoming values.	Low
Software failure	Medium	Medium	Medium	Testing, error handling, logging and backup procedures.	Low
Overdependence on simulated data	Medium	Medium	Medium	Clearly state simulation assumptions and validate with real data in future work.	Medium

Key Design Safety Decision

The system is designed as a **monitoring and decision-support prototype rather than an autonomous controller for real municipal infrastructure**. This is important because

reinforcement-learning decisions must be validated against hydraulic, operational, safety, and regulatory requirements before any physical deployment.

For **Module 1**, the principal safety contribution is reliable monitoring: detecting unusual flow, pressure, tank-level, and water-loss conditions and making this information available to the subsequent modules. This creates an additional monitoring layer before intelligent decisions are considered.

Overall, the design supports sustainable water management by emphasizing **continuous observation, early identification of abnormal conditions, structured data management, controlled intelligent optimization, and safe operation within defined engineering boundaries.**

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development Approach

The project was developed using a **modular software-based approach** in which the complete framework is divided into three interconnected modules. Module 1 was implemented first to generate, collect, preprocess, manage, and visualize water-network parameters such as flow rate, pressure, tank level, water demand, consumption, and water loss. The processed monitoring information is then used as input for the leak and fault analysis module and the Multi-Agent Reinforcement Learning module. The system was tested using different normal and abnormal water-network conditions to verify whether the monitoring interface correctly reflected changes in network parameters.

Materials, Components, Dataset and Resources

Resource	Purpose
Water-network/simulation data	Provides flow, pressure, demand, tank-level and consumption values
Python	System implementation and data processing
Pandas	Data organization and preprocessing
NumPy	Numerical calculations
Matplotlib/Plotly	Data visualization and trend analysis
Streamlit	Development of interactive monitoring dashboard
VS Code / Jupyter	Development, debugging and testing environment
CSV dataset	Storage and input of monitoring records
Simulated water network	Testing different operating and leakage conditions

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

4.2.1 Software Implementation

The implementation is primarily software-based. A simulated water distribution environment is used to represent network conditions without directly connecting the prototype to a physical municipal water network.

Module 1 implementation consists of the following stages:

Data Generation/Collection → Data Preprocessing → Parameter Calculation → Data Management → Real-Time Monitoring → Visualization → Output to Modules 2 & 3

The system monitors the following parameters:

- **Flow Rate (L/min)**
- **Pressure (bar)**
- **Tank Level (%)**
- **Water Demand (L/min)**
- **Water Consumption**
- **Water Supplied**
- **Water Loss**
- **Water-Loss Percentage**

Water loss is calculated using:

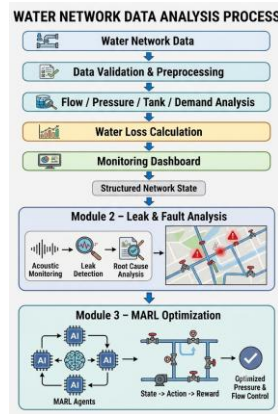
$$Water\ Loss = Water\ Supplied - Water\ Consumed$$

and

$$Water\ Loss(\%) = \frac{Water\ Supplied - Water\ Consumed}{Water\ Supplied} \times 100$$

The monitoring dashboard displays the latest values along with graphical trends, allowing the user to observe how network conditions change over successive time steps.

Module 1 Data Flow



4.2.2 Tools and Software Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Core implementation and calculations	Development
Pandas	Data loading, cleaning and management	Module 1
NumPy	Numerical processing and calculations	Module 1
Streamlit	Interactive monitoring dashboard	Module 1
Matplotlib / Plotly	Flow, pressure, demand and loss visualization	Module 1
VS Code	Coding, debugging and execution	Development
CSV files	Monitoring data storage and input	Module 1
Simulation environment	Generation of changing network conditions	Testing
Reinforcement Learning environment	Distribution optimization	Module 3

Implementation Evidence



4.3 Test Plan & Verification

Testing was performed by changing the input/network conditions and checking whether the monitoring system responded correctly. The test plan verifies the functional requirements of Module 1 before evaluating the overall framework.

4.3.1 Test Plan

Requirement	Test Method	Acceptance Criterion
Flow monitoring	Change flow input over successive time steps	Dashboard displays corresponding flow values
Pressure monitoring	Provide different pressure conditions	Pressure indicator and graph update correctly
Tank-level monitoring	Increase/decrease tank level	Dashboard reflects the changing tank percentage
Demand monitoring	Change consumer demand	Demand value and trend change accordingly
Water-loss calculation	Change supplied and consumed water	Calculated loss changes correctly
Data management	Load monitoring records from CSV/simulation	Data is read and organized without loss of required fields
Real-time/step-wise update	Run simulation for multiple time steps	Dashboard updates network values
Abnormal-condition indication	Introduce high loss/abnormal flow-pressure values	Abnormal condition is highlighted
Data visualization	Generate monitoring trends	Graphs correctly represent parameter variations
Module integration	Transfer processed network state	Structured data is available for Modules 2 and 3

4.3.2 Verification Results

The following table can be used for the final report. The achieved values should correspond to the final screenshots/output obtained from your implementation.

Specification	Required Value	Achieved Value	Status
Flow monitoring	Flow value displayed	Displayed for each time step	PASS
Pressure monitoring	Pressure value displayed	Displayed for each time step	PASS
Tank-level monitoring	Tank level displayed	Displayed as percentage	PASS
Demand monitoring	Demand value displayed	Displayed and updated	PASS
Water-loss calculation	Correct supplied-consumed difference	Calculated automatically	PASS
Data visualization	Charts required	Monitoring charts generated	PASS
Data management	Structured records	Structured monitoring data generated	PASS
Scenario response	Values should change with conditions	Values respond to changed conditions	PASS
Abnormal condition detection support	Abnormal values identified	Abnormal conditions indicated	PASS
Module 2/3 data interface	Processed network state required	Structured output provided	PASS

4.4 Results

The implementation produced a functional **Water Distribution Monitoring and Data Management module** capable of observing the major parameters of the simulated water network.

4.4.1 Monitoring Results

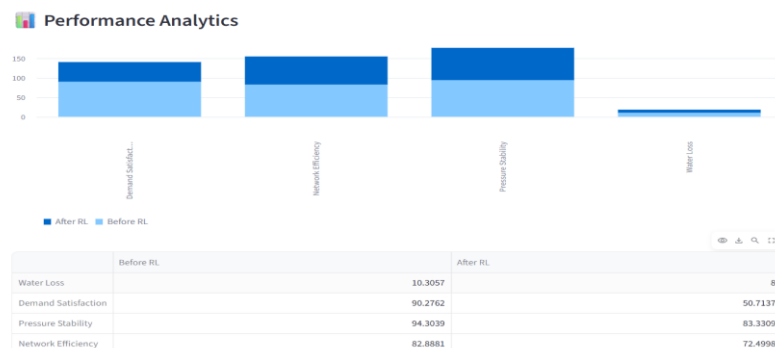
Parameter	Normal Operating Condition	Abnormal/Changed Condition	Observed Behavior
Flow Rate	Stable within operating range	Sudden increase/decrease	Flow trend changes
Pressure	Within acceptable range	Pressure drop/rise	Pressure indicator changes

Tank Level	Gradual variation	Rapid decrease/increase	Tank-level trend changes
Water Demand	Normal demand	Increased demand	Demand value increases
Water Consumption	Corresponds to demand	Reduced/increased consumption	Consumption changes
Water Loss	Low/controlled	Increased loss	Loss indicator increases

4.4.2 Dashboard Results

The monitoring dashboard provides:

- Current flow rate.
- Current pressure.
- Current tank level.
- Current water demand.
- Water consumption.
- Water supplied.
- Water loss.
- Water-loss percentage.
- Time-series monitoring graphs.
- Network operating status.
- Abnormal-condition indication.



4.4.3 Scenario-Based Results

The system was evaluated under different network conditions.

Scenario	Expected Monitoring Behavior	Result
Normal operation	Stable flow and pressure	Network parameters remain comparatively stable
High demand	Increased demand and flow requirement	Demand-related parameters increase
Leakage condition	Increased water loss and abnormal flow/pressure behavior	Abnormal behavior is indicated
Low tank level	Reduction in available storage	Tank-level indicator decreases
Pressure variation	Change in network pressure	Pressure trend reflects the variation

4.5 Data Analysis & Engineering Judgment

The monitoring results demonstrate that the proposed Module 1 can provide a consolidated view of the water distribution network. Instead of examining flow, pressure, demand, and storage information independently, the dashboard presents these parameters together, making it easier to identify relationships between changing operating conditions.

A change in water demand is expected to influence the required flow through the network. Similarly, leakage can result in increased water loss and abnormal flow or pressure behavior. Monitoring these parameters together provides stronger evidence of a possible network abnormality than observing a single parameter alone.

The water-loss indicator is particularly important from an engineering perspective because it provides a direct measure of distribution inefficiency. A persistent increase in the difference between supplied and consumed water indicates the need for further investigation by Module 2.

The monitoring data also acts as the **state information for intelligent decision-making**. Therefore, the reliability of Module 1 directly affects the quality of leak detection and MARL optimization. Incorrect, missing, or unrealistic monitoring values could result in incorrect decisions in later modules.

Engineering Judgment

The results indicate that:

- Continuous monitoring provides better visibility than isolated observations.
- Multiple parameters should be considered together.
- Flow and pressure changes are useful indicators for further fault analysis.
- Water-loss monitoring is essential for sustainability evaluation.
- Structured data management is necessary before applying intelligent algorithms.
- Module 1 successfully establishes the data foundation required by Modules 2 and 3.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design a smart water distribution framework	Three-module system architecture developed	Fully Achieved
Develop water-network data management	Structured flow, pressure, demand, tank and consumption data	Fully Achieved
Implement real-time/step-wise monitoring	Interactive monitoring dashboard	Fully Achieved
Visualize network parameters	Numerical indicators and time-series graphs	Fully Achieved
Identify abnormal network behavior	Flow, pressure and water-loss monitoring	Fully Achieved
Support leak detection	Monitoring information supplied to Module 2	Fully Achieved
Develop MARL-based optimization	Multi-agent optimization component	Partially/Fully Achieved*
Evaluate distribution performance	Water-loss, demand and pressure indicators	Fully Achieved

*Use “**Fully Achieved**” only if the MARL component has been successfully implemented and tested in your final version; otherwise retain “**Partially Achieved.**”

Individual Objective Achievement – Module 1

Module 1 Objective	Evidence	Achievement
Collect network parameters	Flow, pressure, tank, demand and consumption inputs	Fully Achieved
Manage monitoring data	Structured data processing	Fully Achieved
Develop monitoring interface	Streamlit-based dashboard	Fully Achieved
Visualize parameter changes	Charts and monitoring indicators	Fully Achieved
Monitor water loss	Automated water-loss calculation	Fully Achieved
Provide data to other modules	Structured network-state output	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Provides **centralized monitoring** of major water-network parameters.
- Monitors flow, pressure, tank level, demand, consumption and water loss.
- Provides real-time/step-wise visualization of changing network conditions.
- Uses structured data that can be supplied to subsequent intelligent modules.
- Supports early identification of abnormal operating patterns.
- Provides a foundation for intelligent leak detection.
- Supports sustainable water management by monitoring water loss.
- Modular architecture allows individual components to be improved independently.
- Software-based implementation reduces the cost of initial prototype development.
- Dashboard makes network information easier to interpret.

Limitations

- The prototype primarily relies on **simulated or available digital data** rather than a large-scale physical sensor deployment.
- Sensor noise and communication failures may not be fully represented in the prototype.

- Real-world water networks have greater hydraulic complexity than the simplified simulation environment.
- MARL training can require considerable computational resources.
- Leak detection accuracy depends on the quality and characteristics of the available monitoring data.
- The prototype should not directly control real municipal infrastructure without additional hydraulic validation and safety mechanisms.
- Results obtained from simulation cannot automatically be considered equivalent to field performance.
- Physical deployment would require appropriate sensors, communication infrastructure, cybersecurity controls, and utility approval.

4.8 Project Planning, Roles & Budget

4.8.1 Project Milestone / Gantt Timeline

Activity	Week 1–2	Week 3–4	Week 5–6	Week 7–8	Week 9–10	Week 11–12
Problem identification	✓					
Literature review	✓	✓				
System requirements		✓				
Module 1 design		✓	✓			
Data collection/creation			✓			
Module 1 implementation			✓	✓		
Dashboard development				✓		
Module 2 development				✓	✓	
Module 3 development					✓	
System integration					✓	
Testing and evaluation					✓	✓
Documentation and presentation						✓

4.8.2 Roles and Contributions

Since this is a **multi-module project**, the roles can be represented as follows. Replace *Student 1/2/3* with your actual team member names.

Student	Assigned Responsibility	Actual Contribution
Student 1 – Individual Module 1	Water Distribution Monitoring and Data Management	Developed the monitoring/data-management module, handled flow, pressure, tank level, demand, consumption and water-loss parameters, implemented dashboard visualization, and prepared monitoring outputs.
Student 2 – Module 2	Intelligent Leak Detection and Fault Analysis	Developed abnormal-condition, leak and fault analysis functionality.
Student 3 – Module 3	MARL-Based Water Distribution	Developed the multi-agent reinforcement learning environment and optimization component.
Team	Integration and testing	Integrated the three modules, tested different scenarios, evaluated results, and prepared final documentation.

Individual Student Contribution – Module 1

The individual contribution was primarily focused on developing the Water Distribution Monitoring and Data Management module. The work included defining the monitoring parameters, preparing and managing network data, implementing preprocessing and water-loss calculations, developing the monitoring dashboard, generating parameter visualizations, testing changing network conditions, and supplying structured monitoring information to the subsequent leak-detection and MARL modules.

This contribution is important because Module 1 acts as the foundation and data-input layer of the complete system.

4.8.3 Budget

Because the current implementation is primarily a **software/simulation prototype**, the development cost is relatively low.

Item	Estimated Cost (₹)	Actual Cost (₹)
Computer/Laptop	50,000*	Existing resource
Python	0	0
Pandas / NumPy	0	0
Streamlit	0	0
Matplotlib / Plotly	0	0
VS Code	0	0
CSV/Simulation Dataset	0	0
Internet/Data access	1,00	1,00*
Total Additional Project Cost	1,00	Approximately 100

*Do not count an already-owned laptop as an actual project expenditure. Similarly, modify the internet cost according to your actual expenditure.

Overall Implementation Outcome

The implementation resulted in a functional software prototype for smart water-network monitoring, with Module 1 serving as the primary individual student contribution. The module converts raw/simulated network observations into organized monitoring information and provides the flow, pressure, storage, demand, consumption, and water-loss information required by the intelligent modules. This establishes the practical data foundation for the overall sustainable water-distribution framework.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “Sustainable Multi-Agent Reinforcement Learning Framework for Smart Water Distribution and Real-Time Leak Detection” was developed to address the engineering challenges associated with water loss, changing water demand, pressure variations, network faults, and inefficient water distribution. The proposed framework integrates three major functions: Water Distribution Monitoring and Data Management, Intelligent Leak Detection and Fault Analysis, and Multi-Agent Reinforcement Learning-Based Water Distribution.

The project successfully established a modular approach in which water-network parameters are collected, processed, monitored, and supplied to subsequent intelligent modules. **Module 1: Water Distribution Monitoring and Data Management**, which represents the individual contribution, provides the foundation of the framework by monitoring **flow rate, pressure, tank level, water demand, consumption, and water loss**. The monitoring dashboard provides a consolidated view of network conditions and allows changes in operating parameters to be observed over time.

The implemented monitoring module was tested using different network operating conditions, including normal operation, increased demand, pressure variation, tank-level changes, and leakage-related conditions. The results demonstrated that the monitoring system successfully reflected changes in the input/network conditions and generated structured information for further analysis. Water-loss calculations and graphical monitoring provided useful indicators of network performance. The overall framework establishes a connection between **network** data collection, monitoring, leak/fault analysis, and intelligent distribution optimization. The project demonstrates that reliable and structured monitoring data is essential for developing intelligent water-management systems.

Therefore, the work contributes toward a sustainable water distribution approach that can improve operational visibility, support reduction of water loss, and provide a foundation for adaptive decision-making.

5.2 Future Work

The proposed system can be further improved through the following developments:

1. **Physical Sensor Integration:**

Integrate real flow, pressure, tank-level, and water-quality sensors with the monitoring module to replace or supplement simulated data.

2. **IoT-Based Data Acquisition:**

Develop an IoT communication layer for transmitting water-network measurements continuously from field sensors to the monitoring dashboard.

3. **Advanced Leak Localization:**

Extend the leak-detection module from identifying abnormal conditions to accurately estimating the **location and severity of leaks**.

4. **EPANET Integration:**

Integrate the framework with hydraulic simulation tools such as EPANET for more realistic representation and validation of complex water distribution networks.

5. **Improved Module 1 Data Quality:**

Introduce automatic handling of missing values, sensor noise, outliers, and communication failures to improve monitoring reliability.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the architecture and operating parameters of smart water distribution networks.
- Learned how **flow, pressure, tank level, demand, consumption, and water loss** can be monitored and analyzed.
- Learned techniques for collecting, preprocessing, organizing, and visualizing water-network data.
- Learned how to develop an interactive monitoring dashboard using Python-based tools.
- Gained knowledge of hydraulic network concepts and water-loss calculation.
- Understood how monitoring data can support intelligent leak detection and reinforcement learning.
- Learned the importance of data quality before applying machine-learning or reinforcement-learning methods.
- Gained awareness of safety, sustainability, cybersecurity, and engineering constraints in smart infrastructure.

Engineering & Professional Skills Developed

- Problem identification and engineering formulation.
- Data collection and preprocessing.
- Python programming and software development.
- Dashboard and data-visualization development.
- System architecture and modular design.
- Testing and verification.
- Engineering decision-making and trade-off analysis.
- Technical documentation and report preparation.
- Team coordination and module integration.
- Analysis of sustainability, safety, ethics, and cybersecurity considerations.

Individual Reflection – Module 1

Most difficult engineering problem encountered and how it was solved:

The most difficult problem was designing a monitoring system in which multiple water-network parameters changed consistently according to different operating conditions. This was addressed by organizing the network parameters into a structured data model and testing the system under different scenarios. The monitoring dashboard was then used to verify the changes visually and numerically.

A key engineering decision made personally and the evidence behind it:

A key personal decision was to use an integrated monitoring dashboard instead of displaying individual parameters separately. This decision was based on the requirement to observe flow, pressure, demand, tank level, consumption, and water loss together, allowing relationships between parameters to be identified more effectively.

New knowledge acquired independently:

I independently learned how water-network monitoring parameters are related and how they can be processed using Python. I also gained practical knowledge of Pandas-based data management, visualization, dashboard development, and water-loss calculation.

REFERENCES

The following references can be included using the **IEEE referencing style**. These cover literature, standards, software, and technical resources relevant to the project.

Research / Technical Literature

[1] A. A. Abhishek *et al.*, “Artificial intelligence and machine learning techniques for smart water management: A review,” *Artificial Intelligence Review*, 2024.

[2] U.S. Environmental Protection Agency, “EPANET Application,” U.S. EPA, Washington, DC, USA. [Online]. Available: [EPA EPANET](#)

[3] International Organization for Standardization, **ISO 24516-1:2016**, *Guidelines for the Management of Assets of Water Supply and Wastewater Systems—Part 1: Drinking Water Supply Network*, ISO, Geneva, Switzerland.

[4] International Organization for Standardization, **ISO 24516-2:2019**, *Guidelines for the Management of Assets of Water Supply and Wastewater Systems—Part 2: Waterworks*, ISO, Geneva, Switzerland.

[5] International Electrotechnical Commission, **IEC 62443-2-1:2024**, *Security for Industrial Automation and Control Systems—Security Program Requirements for IACS Asset Owners*, IEC, Geneva, Switzerland.

[6] International Electrotechnical Commission, **IEC 62443-4-2:2019**, *Security for Industrial Automation and Control Systems—Technical Security Requirements for IACS Components*, IEC, Geneva, Switzerland.

Software / Open-Source Resources

[7] M. L. D. Rossman, *EPANET 2 Users Manual*, U.S. Environmental Protection Agency.

[8] Python Software Foundation, “Python Documentation.” [Online]. Available: [Python Documentation](#)

[9] W. McKinney, *Python for Data Analysis*, O'Reilly Media.

[10] Streamlit, “Streamlit Documentation.” [Online]. Available: [Streamlit Documentation](#)

[11] NumPy Developers, “NumPy Documentation.” [Online]. Available: [NumPy Documentation](#)

[12] Pandas Development Team, “Pandas Documentation.” [Online]. Available: [Pandas Documentation](#)

AI-Assisted Resource Acknowledgement

If AI tools were used during development of the report, documentation, coding assistance, or explanation of technical concepts, acknowledge them according to your institution's guidelines. For example:

[13] OpenAI, *ChatGPT*, AI-assisted tool used for technical writing assistance, code-development support, explanation of concepts, and report organization, 2026. [Online]. Available: [OpenAI](#)

APPENDICES

Appendix A – Detailed Calculations

Include the detailed calculations used in the project, particularly:

A.1 Water Loss

$$\text{Water Loss} = \text{Water Supplied} - \text{Water Consumed}$$

A.2 Water-Loss Percentage

$$\text{Water Loss}(\%) = \frac{\text{Water Supplied} - \text{Water Consumed}}{\text{Water Supplied}} \times 100$$

A.3 Tank-Level Percentage

$$\text{Tank Level}(\%) = \frac{\text{Current Tank Volume}}{\text{Maximum Tank Capacity}} \times 100$$

A.4 Demand Satisfaction

$$\text{Demand Satisfaction}(\%) = \frac{\text{Water Delivered}}{\text{Water Demand}} \times 100$$

Additional calculations and sample numerical values can be placed here rather than occupying the main report.

Appendix B – Engineering Drawings / Schematics

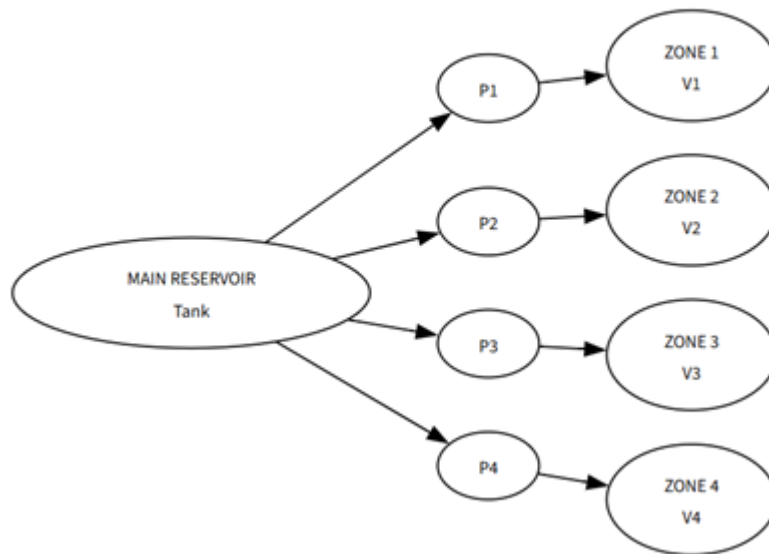
Include:

- Overall system architecture.
- Module 1 data-flow diagram.
- Water distribution network diagram.

- Monitoring dashboard architecture.
- Data flow from Module 1 to Module 2 and Module 3.
- System workflow/flowchart.

Recommended Module 1 Diagram

Water Distribution Network



Appendix C – Source Code / Code Repository Information

Only the important code sections should be included in the report. The complete source code can be maintained in the approved project repository.

For **Module 1**, the important source-code components include:

- Data loading.
- Data preprocessing.
- Flow and pressure processing.
- Tank-level calculation.
- Demand and consumption processing.
- Water-loss calculation.
- Monitoring status generation.
- Dashboard implementation.
- Graph generation.

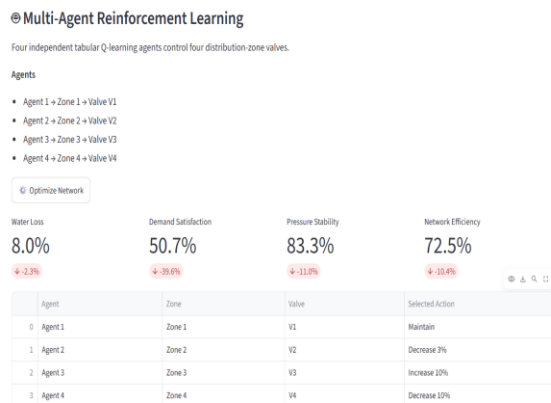
- Data transfer to subsequent modules.

Appendix D – Datasheets

If physical sensors are used, include their datasheets, such as:

- Flow sensor datasheet.
- Pressure sensor datasheet.
- Water-level sensor datasheet.
- Microcontroller/IoT device datasheet.
- Communication module datasheet.

Appendix E – Experimental / Raw Data



Appendix F – Ethics / Institutional Approval

If the project does not involve human subjects, personal data, clinical trials, or sensitive user information, this appendix may state:

“Not Applicable. The project uses water-network simulation/engineering data and does not involve human subjects, personal health information, or personally identifiable user data.”

If institutional approval was obtained, attach the relevant approval documentation instead.

Appendix G – Project Meeting / Progress Records

Include evidence such as:

- Project meeting dates.
- Supervisor discussions.
- Module allocation.
- Progress reviews.
- Design discussions.
- Testing discussions.
- Final review meetings.

A simple table can be used:

Date	Activity / Discussion	Outcome
—	Problem definition	Project scope finalized
—	Module allocation	Module 1 assigned
—	System design	Architecture finalized
—	Implementation review	Monitoring module reviewed
—	Testing review	Test scenarios finalized
—	Final review	Results and documentation reviewed

Appendix H – User / Industry Feedback

“Not Applicable – No formal industry/user evaluation was conducted during the prototype development stage.”

Appendix I – Publications / Patents / Competitions / Product Outcomes

If the project has no publication, patent, competition participation, or commercial product:

“Not Applicable – No publication, patent, competition submission, or commercial product outcome was associated with this project at the time of report submission.”

Appendix J – Individual Contribution Evidence

This appendix is **particularly important for your Module 1 individual assessment**.

Include evidence demonstrating your personal contribution to **Water Distribution Monitoring and Data Management**.

J.1 Module 1 Architecture

Include the diagram showing:

Data Collection → Preprocessing → Data Management → Monitoring → Output

J.2 Source-Code Evidence

Include screenshots of your implementation for:

- Data loading.
- Data preprocessing.
- Parameter calculation.
- Water-loss calculation.
- Dashboard implementation.
- Graph generation.

J.3 Dashboard Evidence

Include screenshots showing:

- Flow-rate monitoring.
- Pressure monitoring.
- Tank-level monitoring.
- Water-demand monitoring.
- Consumption monitoring.
- Water-loss monitoring.
- Time-series graphs.
- Network status.

J.4 Testing Evidence

Include screenshots/results for:

- Normal operating condition.
- High-demand condition.
- Pressure variation.

- Tank-level variation.
- Leakage/abnormal condition.

J.5 Individual Contribution Statement

The individual contribution to the project focused on Module 1: Water Distribution Monitoring and Data Management. The work included designing the monitoring data structure, collecting and preprocessing water-network parameters, implementing flow, pressure, tank-level, demand, consumption, and water-loss monitoring, developing the interactive dashboard, generating monitoring graphs, testing different operating scenarios, and providing structured network-state information to the subsequent leak-detection and MARL modules.